

Create a C++ project called **Surname_studentNumber_Week4Ass4**.

A group of 10 students need to present their programming projects for marks.

1. Declare an array of type string called **arrNames** with the names of 10 students.
Declare another array of type string called **arrPresent** to keep the names of the students in the order that they will present their projects.
Declare an array of type integer called **arrMarks** to keep the marks of the students.
2. Write a function called **displayArray()** to receive the arrNames array and display a numbered list of the content of the array.
3. Example of out if the function is called:

List of names

```
1. Peter
2. Diane
3. George
4. Frank
5. Graig
6. Zane
7. Jacky
8. Mary
9. Elizabeth
10. Sharon
```

4. The order of presenting projects must be generated randomly.

Fill up the **arrPresent** with the names of students from the arrNames array.

As names are selected from the arrNames array the names must be cleared from the arrNames array (replaced by "").

Do this as follows:

Use a for loop as the outer loop to refer to each element of the arrPresent array. Inside the for loop do the following:

Use a do while loop inside the for loop to generate a random value in the range from 0 to 9 to until the name that is found in the arrNames array using the random value is not a blank space.

```
do
{
    index = rand() % 10;
}
while (arrNames[index] == "");
```

Assign the name in the random index position from the arrNames array to the arrPresent using the counter of the for loop.

Assign "" in the random index position in the arrNames to clear the name from arrNames.

NOTE: arrNames should be filled with empty strings once the for loop is done. All the names should then be assigned to arrPresent in random positions in the array.

Call the **displayArray()** function passing the `arrPresent` array to the function to see the shuffled list of names.

Example of shuffled list of names in the `arrPresent` array:

```
List of names
1. Jacky
2. Zane
3. George
4. Diane
5. Sharon
6. Frank
7. Graig
8. Elizabeth
9. Peter
10. Mary
```

NOTE: The list will be different each time when the code is executed due to random selection of names.

5. Write code to allow the user to swap names in the **arrPresent** array. Use a while loop which asks the user whether a swap needs to be done or not. If Y, call the **displayArray()** function to display the numbered list of names from the **arrPresent** array. Ask the user to enter the number of the student on the list who wants to swap. Ask the user to enter the number of the student on the list who is willing to swap. Write code to swap the two names.

Call the **displayArray()** function to show the updated list.

Ask if another swap must be done.

Repeat this process until the user enters n – no more names to be swapped.

How to swap two elements in an array:

Three steps:

```
string temp = arrPresent[index -1];
arrPresent[index -1] = arrPresent[newIndex -1];
arrPresent[newIndex -1] = temp;
```

Example of input and output to swap two names. The program must be able to swap more names until the user enters n.

```

Do you want to swap names (y or n)? y
Enter the number of the name from the list to swap 4
Enter the new position for the name from the list 7
List of names
1. Jacky
2. Zane
3. George
4. Graig
5. Sharon
6. Frank
7. Diane
8. Elizabeth
9. Peter
10. Mary
Do you want to swap names (y or n)? n

```

6. Write code to allow the user to enter the marks of the students as they present their projects using a for loop. Use the arrPresent array to display the name and enter the mark.

Example of input of marks:

```

Do you want to swap names (y or n)? n
Enter the mark for Jacky: 45
Enter the mark for Zane: 76
Enter the mark for George: 90
Enter the mark for Graig: 80
Enter the mark for Sharon: 32
Enter the mark for Frank: 61
Enter the mark for Diane: 88
Enter the mark for Elizabeth: 92
Enter the mark for Peter: 77
Enter the mark for Mary: 69

```

NOTE: The arrPresent array and the arrMarks array are parallel arrays because the elements of the arrays are related.

7. Write a function called **displayMarks()** that will receive the arrPresent array and the arrMarks array and display a numbered list of the names and marks of the students.

Call the **displayMarks()** function from the main function.

8. Write a function called **findBestProject()** that will receive the arrMarks array. Find and return the index value of the best mark in the arrMarks array.

Exmple of display of names and marks:

```

Names and marks
1. Jacky      45
2. Zane       76
3. George     90
4. Graig      80
5. Sharon     32
6. Frank      61
7. Diane      88
8. Elizabeth  92
9. Peter      77
10. Mary     69

```

Call the **findBestProject()** function from the main function and display the name of the student with the best mark and the mark using the index value that was returned by the function..

Example of output:

```
|The student with the highest mark is Elizabeth with a mark of 92
```